

```

#include <iostream>
using namespace std;
double f(double x) {
    return x * x;
}
double trapezoidal(double a, double b, int n) {
    double h = (b - a) / n;
    double sum = 0.5 * (f(a) + f(b));
    for (int i = 1; i < n; i++) {
        sum += f(a + i * h);
    }
    return sum * h;
}
int main() {
    double a, b;
    int n;
    cout << "Enter the lower limit (a): ";
    cin >> a;
    cout << "Enter the upper limit (b): ";
    cin >> b;
    cout << "Enter the number of subintervals (n): ";
    cin >> n;
    double result = trapezoidal(a, b, n);
    cout << "The approximate integral from " << a << " to " << b << " is: " << result << endl;
    return 0;
}

```